STAFF REVIEW · H446 256 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Different instructions have different costs, and decoding, caches and pipelining affect performance.

2. An execution unit capable of carrying out its own instruction stream.

3. 4 minutes. Both cores execute at the same time and each finishes its task after 4 minutes.

4. Final [2,2,6,4], sum=14. Indexes 0 and 2 were changed, not the even-valued elements.

I Do — source worked steps

1. Predict first — Divide 12 independent equal-sized tiles among four available cores.

2. Explain and check — Each core processes three tiles, so ideal completion is around one quarter of serial time only if work and overhead allow.

We Do — guided CHECK

1. Predict first
The blocks can be divided evenly into five independent groups.
Explain and check
Each core receives four blocks; coordination overhead and data movement may affect real runtime.

You Do — source mark schemes

1. [4] Graphics/rasterisation [1]; large-scale matrix operations/AI [1]; many independent similar calculations [2].

2. [6] 24/6=4 chunks per core [2]; serial work limits speedup [2]; synchronisation, memory-bandwidth or scheduling overhead [2].

3. [6] Independent per-frame filter calculations can be parallel [2]; GPU data-parallel execution benefits them [1]; decoding dependencies may constrain speedup [1]; assembly waits for processed frames [1]; host–GPU transfer and memory bandwidth reduce ideal gains [1].

Exit Ticket — source answer

1. False. Highly dependent tasks may not benefit and transfer/launch overhead can dominate.